30-DAY GIT LEARNING SERIES

GIT QUEST

LEVEL 9

git restore and git reset: Oops!

Week 2: Undo and Branch | Absolute Beginner

QUEST PROGRESS: DAY 9 OF 30

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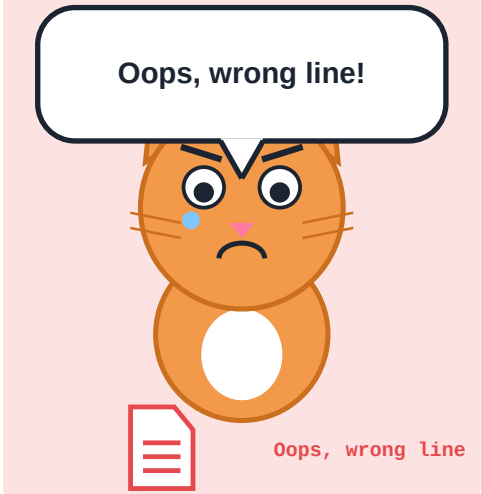
The Story: The Undo Button

Commit the Cat made a mistake. Luckily, Git has four ways out.

1



The mistake

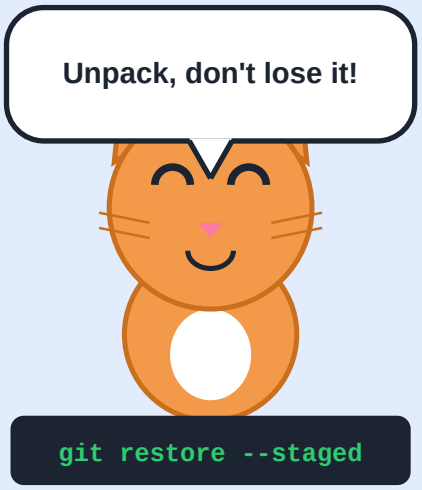
Everyone edits the wrong thing sometimes. The trick is picking the right undo.

2

git restore

Throws away edits to a file and brings back the last saved version.

3



--staged

Takes a file off the packing table. Your edits stay safe on the desk.

4

git reset

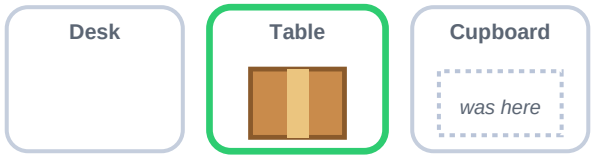
Takes the last commit out of the cupboard. You choose where its changes go.

The 3 Reset Modes

git reset HEAD~1 takes the last commit out of the cupboard. Where do its changes go?

SOFT — Safe

```
git reset --soft HEAD~1
```



Commit undone. Changes stay staged on the packing table.

MIXED (default) — Safe

```
git reset HEAD~1
```

Commit undone. Changes go back to the desk, unstaged.

HARD — DANGER

```
git reset --hard HEAD~1
```

Commit undone. Changes are thrown away!

WHICH UNDO DO I NEED?

I edited a file and want those edits gone	`git restore <file>`	loses edits
I staged the wrong file	`git restore --staged <file>`	safe
I committed too early but want to keep my work	`git reset --soft HEAD~1`	safe
I committed junk and want it gone completely	`git reset --hard HEAD~1`	loses work

Danger zone: restore <file> and reset --hard can't be undone!

Quest Board: Practice the Undo

Finish 4 quests, then collect today's commit.

1 Save a one-line story

```
$ git init -b main undo-lab && cd undo-lab
$ echo "Line 1" > story.txt
$ git add story.txt && git commit -m "Add story"
```

2 Oops! Throw away a bad edit

```
$ echo "Oops, wrong line" >> story.txt
$ git restore story.txt
```

story.txt is back to Line 1 only.

3 Unstage without losing work

```
$ echo "Line 2" >> story.txt
$ git add story.txt
$ git restore --staged story.txt
```

Your edit is kept: it is just back on the desk.

4 Undo a commit (soft)

```
$ git add . && git commit -m "Add line 2"
$ git reset --soft HEAD~1
$ git status
```

Commit gone, change still staged. Try --mixed too. Avoid --hard!

LOOT: TODAY'S COMMIT

COMMIT

days-1-to-9-summary.md

```
docs(day09): add days 1-9 summary and undo notes
```

Recap of Days 1 to 9 plus undo notes.

SAFETY RULES

1. Run git status and git diff first.
2. restore <file> and reset --hard can't be undone for unsaved work.
3. Only reset commits that are still on your computer.

RESCUE TOOL (DAY 26)

```
$ git reflog
```

Lists where HEAD has been, so reset commits can often be rescued.

Boss Fight and Bonus Loot

Cheat sheet, safety ladder, quiz and your next unlock.

BONUS 1: DAY 9 CHEAT SHEET

```
git restore <file>                  Discard edits (DANGER)
git restore --staged <file>         Unstage, keep edits
git restore --source=HEAD~1 <file>  Get an older version
git reset --soft HEAD~1             Undo commit, keep staged
git reset HEAD~1                    Undo commit, keep on desk
git reset --hard HEAD~1             Undo commit, delete work
```

BONUS 2: LOOK BEFORE YOU UNDO

```
git status     Which area is each file in?
git diff       What exactly will I lose?
```

BONUS 3: THE SAFETY LADDER (SAFEST FIRST)

```
1. git restore --staged <file>
2. git reset --soft HEAD~1
3. git reset HEAD~1   (mixed)
4. git restore <file> / git reset --hard   (last resort)
```

BOSS FIGHT: QUICK QUIZ

1. Which command throws away unstaged edits in a file?
2. What is the difference between reset --soft and reset --hard?
3. Which is safe: restore --staged or plain restore?

Answers: 1) git restore <file> 2) --soft keeps changes staged, --hard deletes them 3) restore --staged is safe

CHALLENGE

Make two commits. Undo the last one with --soft, then commit again with a better message.

LEVEL 10: git commit --amend

Unlocks tomorrow. Fix your last commit in one command.

LOCKED